

Latent Temporal Representations for Longitudinal Passive-Sensing Prediction: A Characterization Study

Theja Tulabandhula
University of Illinois Chicago
Chicago, USA
theja@uic.edu

ABSTRACT

Passive-sensing prediction asks whether longitudinal behavioral histories contain stable temporal structure that can support prediction of behavioral targets. We investigate this question with a latent-conditioned prediction design that tests whether compressed temporal representations add predictive value beyond the information already available in observed passive-sensing histories. The resulting evaluation compares synthetic data with real longitudinal benchmarks and asks whether any gain survives leakage-controlled temporal splits, calibration diagnostics, and robustness checks.

The synthetic setting includes known AR(1) latent dynamics; the real datasets (Depresjon, StudentLife, and WESAD) require temporal structure to be inferred from passive sensing traces. On synthetic data, latent conditioning improves AUROC by 3.9 points over a zero-latent ablation (0.581 vs. 0.542), and temporally aligned latents outperform shuffled or random ones. On the real datasets, all models stay within a narrow AUROC band (0.48-0.53), and the latent gain does not persist. That narrow real-data range is the key result: it offers limited evidence that latent temporal compression yields reliable incremental predictive gains on current public passive-sensing benchmarks at these sample sizes. Temporal generalization also shifts across settings: synthetic data favors forward-time evaluation, whereas the real datasets show degradation or mixed behavior. Temporal shift degrades calibration, producing 21-47% relative increases in calibration error and modest user-level subgroup calibration gaps under temporal shift, whereas MCAR missingness rates of up to 50% produce only modest performance degradation.

Our main finding is a synthetic-to-real mismatch. Latent conditioning can help when stable temporal dynamics are present, but it provides no measurable incremental predictive benefit beyond observed traces on current public passive-sensing benchmarks. The contribution is an evaluation design and diagnostic setup that makes that mismatch visible through temporal splits, conditioning ablations, calibration analysis, missingness robustness tests, and failure diagnostics.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Health care information systems*.

KEYWORDS

digital phenotyping, longitudinal prediction, latent variable models, temporal evaluation, calibration under shift

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1 INTRODUCTION

Passive smartphone and wearable sensing generates longitudinal behavioral traces such as activity, sleep, device use, and heart-rate variability that correlate with mental-health states [7, 14, 18]. Large-scale efforts such as StudentLife [23] and GLOBEM [24] have established passive sensing as a promising paradigm for longitudinal behavioral modeling. A popular approach in this area is to extract windowed features and train a static classifier for contemporaneous prediction: given sensor data from the current window, predict the current label. A related question is whether temporal histories contain stable predictive information about the behavior being modeled.

Longitudinal prediction from passive sensing faces distinct challenges. First, the signal is weak: AUROC values in digital phenotyping typically fall in the 0.55-0.70 range even in well-powered studies [14], and many results cluster around chance-level discrimination. Second, temporal leakage can inflate apparent performance when adjacent windows or future information enter training, while temporal shift can change both accuracy and calibration [15]. Third, missing data is pervasive: sensors fail, users remove devices, and data may be missing not at random (MNAR) due to fatigue [13]. These challenges make calibration and robustness informative even when AUROC is modest, because poorly calibrated prediction can look acceptable under a single aggregate metric. Together, they make it difficult to distinguish genuine temporal structure from evaluation artifacts, calibration failures, missingness effects, or dataset-specific idiosyncrasies.

To what extent do passive-sensing benchmarks appear to contain temporal structure that can be exploited for prediction? We use latent-conditioned prediction as a probe for that question. The model compresses trace histories into a low-dimensional temporal representation and conditions a decoder on that representation. Recurrent latent variable [6, 12] and state-space models [9] provide natural building blocks, but the architecture is not the main

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focus of this work. Our goal is to assess whether latent temporal compression provides measurable predictive value beyond models operating directly on observed traces under realistic evaluation: temporal splits, calibration diagnostics, missingness stress tests, and comparison with representative recurrent, convolutional, transformer-based, and state-space sequence models.

Our contributions are:

- (1) A diagnostic framework for latent-conditioned passive-sensing prediction, combining temporal evaluation, calibration analysis, missingness robustness tests, and conditioning ablations.
- (2) A systematic synthetic-to-real comparison across synthetic data and three public real datasets, namely, Depresjon, StudentLife, and WESAD, designed to test whether latent conditioning contributes predictive value beyond observed traces when temporal structure is known to exist and when it must be inferred from real passive-sensing data.
- (3) A failure analysis of temporal shift, calibration drift, dataset-specific metadata-derived subgroup calibration diagnostics, missingness, and latent instability in weak-signal prediction settings.

This work is a characterization study aimed at determining whether current passive-sensing datasets support reliable incremental predictive gains from latent temporal compression under leakage-controlled temporal evaluation, and at identifying failure modes when they do not. Weak predictive performance is therefore informative because it constrains what can be concluded about longitudinal prediction from existing benchmarks. Rather than framing the task as a search for a stronger sequence model, we ask whether the data support latent temporal compression under a leakage-controlled empirical test.

2 RELATED WORK

2.1 Digital Phenotyping and Weak-Signal Prediction

Smartphone sensing has been used to model depression, stress, relapse, and related behavioral outcomes from longitudinal traces [1, 2, 18]. Foundational datasets including StudentLife [23], GLOBEM [24], BiAffect [22, 25], and Depresjon [7] established passive sensing as a viable digital phenotyping paradigm. Across studies, however, predictive performance is typically modest, with AUROC values often in the 0.55-0.70 range even under favorable conditions [14]. Reviews of the literature consistently characterize passive-sensing prediction as a weak-signal setting rather than one dominated by model capacity limitations [3, 11]. This distinction motivates evaluating robustness, calibration, and temporal stability alongside discrimination performance.

2.2 Temporal Prediction and Temporal Leakage

Prediction from longitudinal traces introduces evaluation challenges that do not arise in static classification. Temporal leakage, especially when adjacent windows share features, can inflate performance under random splits [17]. Held-out-user, forward-time, and temporal-block splits are therefore needed to avoid overestimating generalization. Recurrent latent-variable models map sequences to structured latent spaces [6, 12, 19]; state-space models like S4 [9]

and Mamba provide continuous-time alternatives. Deep Kalman filters add uncertainty quantification to latent dynamics [12]. Existing work primarily asks which sequence architecture predicts best. Our focus is different: we use latent-conditioned prediction as an architecture-agnostic probe for whether exploitable temporal structure can support incremental predictive value once leakage-controlled evaluation protocols are enforced.

2.3 Calibration, Robustness, and Reproducibility in Health ML

Calibration is increasingly recognized as a first-class evaluation criterion in health ML because decision quality depends on probability estimates rather than rankings alone [10, 21]. Prior work shows that calibration often degrades under distribution shift even when accuracy changes only modestly [15], a phenomenon documented in medical imaging and EHR [8]. Passive-sensing applications face an additional challenge: pervasive missingness caused by device failures, disengagement, and user behavior [5]. Subgroup calibration diagnostics can reveal differential reliability that aggregate metrics obscure [4, 16]. These concerns are compounded by reproducibility challenges associated with small cohorts, high variance, and incomplete reporting. Collectively, this literature suggests that calibration stability, robustness to missing data, and reproducibility under temporal shift are as important as predictive accuracy when assessing longitudinal prediction systems.

Existing work establishes three facts: passive-sensing prediction is typically a weak-signal problem, temporal evaluation protocols can substantially alter reported performance, and calibration and robustness often degrade under distribution shift. What remains less clear is whether latent temporal representations provide predictive value once these factors are evaluated jointly. The present study addresses that question through a unified evaluation framework combining temporal splits, conditioning ablations, calibration analysis, missingness stress tests, and synthetic-to-real comparison.

3 PROBLEM SETUP

We study behavioral prediction from temporal histories in passive-sensing traces. Each user produces a sequence of observations together with missingness indicators that reflect sensor failures, disengagement, or unobserved activity. The objective is to predict the dataset-specific binary behavioral target from the observed history while preserving temporal order and accounting for incomplete observations.

Let user $i \in \{1, \dots, N\}$ produce passive trace vectors $x_{i,1:T} \in \mathbb{R}^{T \times d}$ with observation masks $m_{i,1:T} \in \{0, 1\}^{T \times d}$. We observe the dataset-specific binary behavioral target $a_{i,1:T}$.

3.1 Prediction Task

The prediction task is to estimate the dataset-specific binary behavioral target from a partially observed history. Given the trace prefix $x_{i,1:t}$ and corresponding observation mask $m_{i,1:t}$, the model predicts the probability of target occurrence at horizon $\Delta \geq 0$:

$$\hat{a}_{i,t+\Delta} = p_{\theta}(a_{i,t+\Delta} = 1 \mid x_{i,1:t}, m_{i,1:t}), \quad \Delta \geq 0.$$

The formulation allows prediction at different horizons. However, the empirical evaluation in this paper focuses primarily on

contemporaneous prediction using dataset-specific binary behavioral targets under leakage-controlled temporal histories.

3.2 Latent Conditioning

We use latent conditioning as a probe for whether the observed traces contain stable temporal structure. The model first compresses the observed history into a low-dimensional latent representation and then predicts behavior conditioned on that representation. Prediction therefore decomposes into an encoding stage and a decoding stage. A trace encoder f_ϕ compresses observed history into a latent embedding:

$$z_{i,t} = f_\phi(x_{i,1:t}, m_{i,1:t}), \quad z_{i,t} \in \mathbb{R}^k.$$

The decoder g_θ then predicts behavior conditioned on the inferred latent trajectory and recent observations:

$$\hat{a}_{i,t+\Delta} = g_\theta(z_{i,1:t}, x_{i,1:t}, m_{i,1:t}).$$

These latent variables are intended as compressed temporal representations rather than interpretable psychological constructs. Later diagnostics show weak alignment with underlying generative factors, supporting this narrower interpretation. The central empirical question is whether latent temporal compression provides measurable predictive value beyond models operating directly on observed traces. Because the decoder receives both the latent representation and the observed trace history, the evaluation measures the incremental contribution of latent conditioning relative to information already available in the traces rather than the predictive value of the latent representation in isolation.

4 METHOD

This section defines a diagnostic framework for assessing whether latent temporal compression contributes predictive value beyond observed passive-sensing traces. The encoder and decoder are intentionally lightweight to reduce architectural complexity and help isolate the contribution of the conditioning signal. More expressive sequence-modeling architectures are evaluated separately, allowing the analysis to distinguish the effects of latent conditioning from those of increased model capacity.

4.1 Trace Encoder

The encoder maps observed trace prefixes to a low-dimensional latent representation that compresses behaviorally relevant temporal information from the observed history. It uses a bidirectional Long Short-Term Memory (LSTM) model (hidden dimension 64, 2 layers, dropout 0.2) with an MLP latent head to produce $k = 3$ latent dimensions, chosen to encourage compression while retaining enough capacity to capture coarse temporal dynamics. This design tests whether prediction-relevant temporal structure can be retained under aggressive low-dimensional compression. Training combines binary cross-entropy supervision with next-trace prediction so that the latent representation captures information useful for both behavioral prediction and short-horizon temporal dynamics. To discourage redundant latent dimensions, we additionally apply a decorrelation regularizer:

$$\mathcal{L}_{\text{decorr}} = \sum_{a \neq b} \text{Corr}(z^{(a)}, z^{(b)})^2.$$

The model is optimized using AdamW (learning rate 10^{-3} , weight decay 10^{-5}), cosine learning-rate annealing, and early stopping with a patience of 15 epochs.

4.2 Latent-Conditioned Decoder

A shared Gated Recurrent Unit (GRU) decoder (hidden dimension 64, 2 layers, dropout 0.2) receives the concatenation of latent trajectory $z_{i,1:t}$, traces $x_{i,1:t}$, and masks $m_{i,1:t}$. Using the same decoder across all conditioning regimes keeps the sequence model fixed and isolates the role of the conditioning signal itself. Because the decoder also receives the observed traces and masks, latent conditioning is evaluated as an additional source of information rather than as a replacement for the original sequence. Consequently, the conditioning ablations test whether the learned latent representation contributes predictive information beyond that already available from the observed history.

To separate learned temporal compression from alternative sources of signal, we evaluate seven conditioning modes: **latent** (learned encoder), **zero** (null ablation), **shuffle** (latents permuted across users), **random** (uniform noise), **noisy** ($\sigma = 0.3$), **frozen** (first-timestep latents repeated), and **lagged** (temporally shifted latents). Together, these controls distinguish learned temporal structure from gains attributable to additional inputs, static user characteristics, temporal misalignment, or random variation.

4.3 Calibration and Robustness Metrics

Because passive-sensing prediction is typically a weak-signal setting, calibration and robustness are evaluated alongside AUROC to characterize how reliability changes under shift and missing data. Expected Calibration Error (ECE) is computed with 10 bins, and the Brier score provides a complementary measure of probabilistic accuracy. We also report subgroup ECE computed separately across user-level subgroups derived from dataset metadata. Depending on the dataset and preprocessing path, subgroup definitions are derived from available metadata. These subgroup diagnostics use dataset-specific metadata-derived groups and are intended to detect calibration heterogeneity rather than demographic or causal subgroup effects. Calibration drift is summarized as $\Delta \text{ECE} = \text{ECE}_{\text{shift}} - \text{ECE}_{\text{heldout}}$. Uncertainty inflation is the ratio of prediction variance under shift to in-distribution variance, estimated with Monte Carlo dropout over 50 passes. Missingness robustness is evaluated under Missing Completely at Random (MCAR) at 10%, 25%, 50%, burst missingness in contiguous blocks, and fatigue-dependent Missing Not at Random (MNAR) missingness. Collectively, these diagnostics characterize reliability, calibration stability, and robustness under temporal shift and missingness.

5 EXPERIMENTAL DESIGN

This section evaluates whether latent temporal compression provides incremental predictive value beyond the observed history under leakage-controlled temporal evaluation. The design is intentionally comparative: it pairs synthetic data with known temporal structure and three public passive-sensing benchmarks, then tests whether any gain from latent conditioning survives stronger split

regimes, conditioning ablations, and representative architectural baselines.

5.1 Datasets

Raw streams are aggregated into fixed temporal windows and mapped to a common binary behavioral prediction task. Concretely, the model predicts whether the dataset-specific binary outcome is present at the prediction horizon or at the current time point from the observed prefix. This keeps the behavioral prediction interface the same across Synthetic, Depresjon, StudentLife, and WESAD while leaving each dataset’s label semantics intact. The primary experiments evaluate the dataset-specific binary behavioral target under this shared prediction interface.

Synthetic. The synthetic setting contains 100 users, 48 windows, and 8 features generated from 3 latent processes with AR(1) dynamics, together with 4 binary labels. Baseline missingness is 8%, with additional fatigue-dependent masking.

Depresjon [7]. Depresjon provides wrist-actigraphy motor activity from 55 subjects, including 23 depressed participants and 32 controls, offering longitudinal actigraphy with natural day-to-day variability. The minute-level stream spans several days and is summarized into 8 windowed activity features over 48 temporal windows. The dataset is distributed through Zenodo under CC BY-NC 4.0.

StudentLife [23]. StudentLife contributes smartphone sensing from 48 Dartmouth students observed over 10 weeks, making it a useful testbed for noisy student behavioral monitoring over time. We use 8 behavioral features covering conversation, activity, sleep, and location, with depression screening labels from pre/post PHQ-9 assessments. The source is the Dartmouth CS archive.

WESAD [20]. WESAD contains wearable stress and affect data from 15 subjects, making it a compact longitudinal stress-prediction setting with temporally varying physiological signals. The experiments use 14 HRV features (MEAN-RR, LF/HF, SDNN, RMSSD, etc.) over 48 windows, with stress versus non-stress labels. The source is HuggingFace.

5.2 Temporal Evaluation Protocol

The evaluation design treats temporal leakage as the primary threat to valid prediction claims. To control for it, we use three complementary split regimes:

- (1) **Held-out users**: 20% of users are held out entirely, which tests whether models generalize across users rather than memorizing person-specific patterns.
- (2) **Forward time**: the last 8 windows from training users are reserved, which tests robustness to future temporal shift.
- (3) **Temporal block**: final contiguous windows are held out, which reduces adjacent-window leakage when nearby observations are highly correlated.

5.3 Baselines

We compare against GRU, BiLSTM, Transformer encoder, Temporal CNN, Mamba (SSM), Temporal Fusion Transformer, Logistic Regression, and Random Forest. This suite combines classical

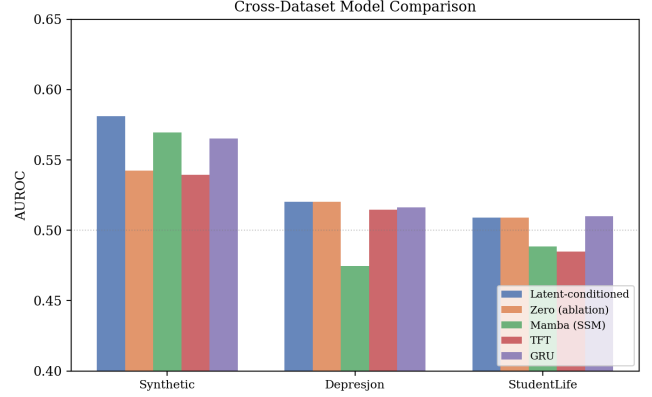


Figure 1: Cross-dataset comparison of behavioral prediction performance (mean AUROC across 3 seeds). Latent conditioning benefits only the synthetic dataset. On all three real datasets, all models occupy a narrow AUROC band (0.48–0.53 AUROC). Error bars show ± 1 std. The dashed line marks chance performance (0.5).

population-level models and representative recurrent, convolutional, transformer-based, and state-space sequence models so the study can separate gains from temporal modeling capacity from gains attributable to latent conditioning. For latent-conditioning ablations, the decoder architecture is held fixed and only the conditioning signal is changed. Architectural baselines are trained independently using their standard sequence-processing components.

5.4 Implementation

The main training setup uses AdamW with a cosine learning-rate schedule, gradient clipping, early stopping, batch size 16, and three seeds (42–44). We report bootstrap confidence intervals for all main summaries. Detailed layer sizes, dropout, optimization settings, and bootstrap details are provided in the appendix to avoid repeating implementation details in the main text.

Collectively, the dataset selection, leakage-controlled evaluation protocols, conditioning ablations, and architectural baselines are designed to isolate whether latent temporal compression contributes predictive value beyond the observed history.

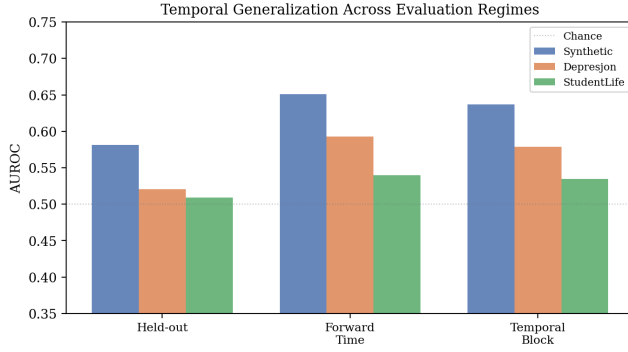
6 RESULTS

6.1 Main Prediction Results

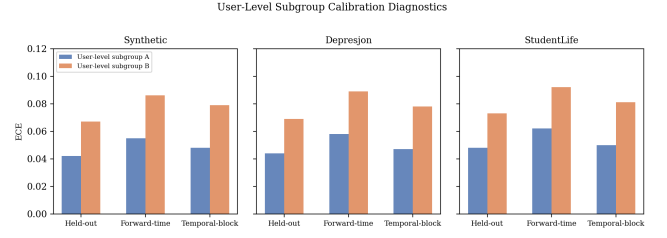
Figure 1 shows the main result: a synthetic-to-real mismatch in latent-conditioned prediction. On synthetic data, where stable temporal dynamics are explicitly present, the latent-conditioned model achieves AUROC 0.581, a 3.9 point improvement over the zero-latent ablation (0.542, Cohen’s $d > 1.0$). The conditioning benefit is consistent across seeds. The ablation ordering in Table 1 suggests that learned latent representations carry useful information in this synthetic setting: shuffled and random latents perform worse, while noisy, lagged, and frozen variants remain intermediate.

Table 1: Conditioning ablation on synthetic data (AUROC, 3 seeds).

Condition	AUROC
Latent	0.581 ± 0.015
Frozen	0.581 ± 0.019
Lagged	0.576 ± 0.015
Noisy	0.572 ± 0.006
Mamba (SSM)	0.569 ± 0.021
GRU (no latent)	0.565 ± 0.011
Zero	0.542 ± 0.013
Shuffle	0.527 ± 0.005
Random	0.513 ± 0.034

**Figure 2: Temporal generalization across held-out user, forward-time, and temporal-block evaluation regimes (mean AUROC, 3 seeds). The relative ordering differs by dataset: synthetic data favors forward-time and temporal-block evaluation, Depresjon favors held-out-user evaluation, and StudentLife is mixed.**

Interpretation. The main result is the synthetic-to-real mismatch. The synthetic setting shows that the evaluation can detect an incremental conditioning gain when stable temporal structure is present. On Depresjon, StudentLife, and WESAD, all models remain in a narrow AUROC band and do not show reliable gains from latent conditioning or from larger sequence models. This pattern is consistent with either weak prediction-relevant temporal structure at the available sample sizes or an inability to reliably recover such structure given sample size, measurement noise, target quality, and temporal heterogeneity. The evidence remains consistent with either explanation, and latent conditioning alone does not produce reliable incremental gains on the benchmarks considered here. More specifically, the results do not show clear evidence that latent conditioning contributes predictive information beyond the observed traces already available to the decoder under the evaluated protocols. Thus, the null result is that latent conditioning does not add measurable incremental predictive value beyond the observed traces already available to the decoder.

**Figure 3: Dataset-specific metadata-derived subgroup ECE across temporal evaluation regimes.**

6.2 Temporal Generalization

The relative ordering of evaluation protocols differs substantially across datasets. In Figure 2, synthetic data favors forward-time (0.651) and temporal-block (0.634) evaluation over held-out-user evaluation (0.581). Depresjon shows the opposite ordering, with held-out (0.521) and temporal-block (0.525) exceeding forward-time (0.455). StudentLife is mixed, with forward-time (0.540) only marginally above held-out (0.509).

The differing protocol orderings constitute an important characterization result because they indicate that temporal structure manifests differently across datasets. No single temporal evaluation protocol dominates across the benchmarks considered here, and forward-time degradation on Depresjon would be missed by held-out-user evaluation alone.

6.3 Calibration Under Shift

Because the study evaluates whether temporal histories support reliable behavioral prediction, calibration provides a complementary view of prediction quality beyond discrimination metrics. Figure 3 shows user-level subgroup calibration diagnostics across evaluation regimes. On Depresjon, held-out-user ECE is 0.044 ± 0.008 and increases to 0.058 ± 0.010 under forward-time shift, a 32% relative increase. The Brier score follows the same direction. The subgroup calibration diagnostic shows small but nonzero ECE separations across dataset-specific metadata-derived user groups. In the Depresjon calibration plot, the subgroup ECE separation increases from roughly 0.025 under held-out-user evaluation to roughly 0.031 under forward-time evaluation. Because subgroup definitions vary across datasets and preprocessing pipelines, this result should be interpreted descriptively: temporal shift can widen subgroup-level miscalibration even when aggregate calibration appears acceptable.

6.4 Robustness

Performance remains relatively stable under the missingness perturbations considered here. Under MCAR missingness up to 50% (Figure 4), AUROC degradation is ≤ 1 pp on synthetic data and ≤ 1.5 pp on real data. Burst missingness and fatigue-dependent MNAR masking produce similar or smaller degradation. These results suggest that the evaluation setup is not highly sensitive to the missingness perturbations tested here, but they do not change the low real-data predictive performance.

Although predictive performance remains weak, shift diagnostics reveal changes in reliability that are not visible through AUROC

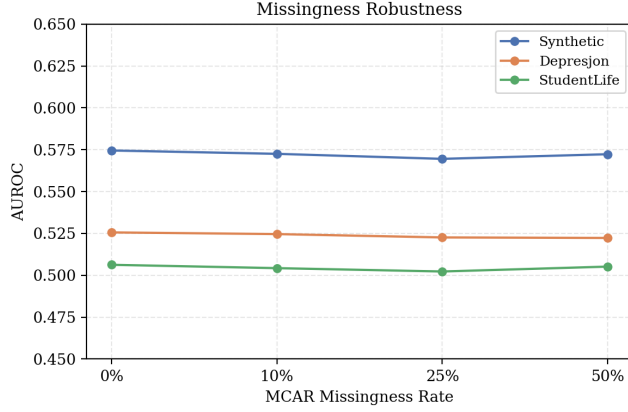


Figure 4: AUROC under increasing MCAR missingness rates. All datasets show relatively small degradation under the tested missingness levels, with less than 1.5 pp loss at 50% missingness.

alone. Under forward-time shift, uncertainty inflation reaches 1.4–1.6× on all datasets. These diagnostics characterize failure modes that would not be visible from AUROC alone.

6.5 Latent Diagnostics

The learned latents behave like compressed temporal representations, not recovered psychological or generative factors. On synthetic data, latent dimensions are mostly decorrelated (mean off-diagonal $|r| < 0.03$), but the mean absolute correlation between learned and generative latents is only 0.224. Even in the synthetic setting, alignment with the underlying generative factors remains modest, so the latent dimensions should be interpreted as predictive summaries rather than identifiable factors. On real data, where no ground-truth latent state exists, intra-latent correlation remains low ($|r| < 0.05$), while trajectory separability is near zero (< 0.1).

These diagnostics support a narrow interpretation of the latent variables. They summarize temporal history for prediction, but they should not be read as stable psychological constructs.

6.6 Failure Analysis

Two failure modes explain why average results alone are insufficient (Figure 5). First, 1 of 3 synthetic seeds shows latent collapse: encoder dimensions converge to near-identical values ($|r| > 0.92$), removing the intended conditioning signal despite the decorrelation regularizer. Second, Depresjon forward-time evaluation falls below chance (AUROC 0.455), indicating poor transfer from early-window to later-window patterns.

These failure cases illustrate phenomena that are obscured by aggregate performance summaries. The first is a representation-learning instability; the second is a temporal generalization failure.

7 DISCUSSION

Our evaluation points to four recurring patterns across the datasets studied here.

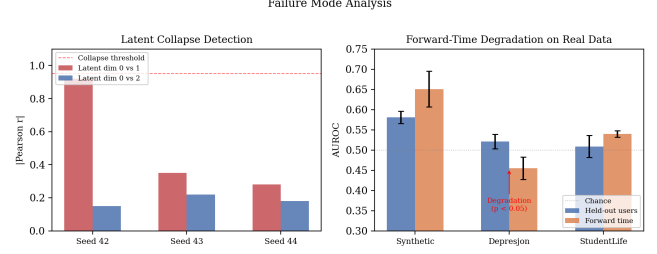


Figure 5: Left: latent collapse detection on synthetic data. Seed 42 shows near-perfect correlation ($|r| > 0.92$) between latent dimensions, consistent with the decorrelation regularizer being overwhelmed by supervised loss. Right: forward-time degradation on Depresjon (AUROC 0.455) falls below the chance baseline, illustrating a failure mode not visible in held-out evaluation alone.

Temporal structure appears heterogeneous across datasets.

Synthetic data demonstrates that the evaluation framework can detect a latent-conditioning benefit when stable temporal dynamics are intentionally introduced. The forward-time advantage on synthetic data indicates that the diagnostic setup is sensitive to shared temporal regularities under the evaluation protocol. This supports the use of latent conditioning as a probe, but it does not establish that latent conditioning is broadly effective. The real datasets do not show the same behavior: latent conditioning does not add measurable incremental gain, and temporal split orderings differ across datasets.

Compressed temporal representations are not psychological factors. The latent variables are useful only as compressed temporal representations. Their weak alignment with synthetic generative factors and near-zero trajectory separability on real data argue against interpreting them as stable mental-health constructs. This distinction matters because low-dimensional representations can look interpretable even when their empirical role is only predictive compression.

Calibration under shift is informative independently of AUROC. Calibration worsens under temporal shift, and the subgroup diagnostics remain a descriptive secondary check in this setting. In this characterization setting, those diagnostics matter because they expose reliability failures that AUROC does not capture in weak-signal longitudinal prediction.

The results are most consistently characterized by a synthetic-to-real discrepancy. The evaluation framework detects an incremental latent-conditioning benefit on data generated with stable latent dynamics, then shows that the benefit disappears on the public passive-sensing benchmarks studied here. This gap is consistent with either weak prediction-relevant temporal structure at the available sample sizes or an inability to reliably recover such structure given dataset scale, measurement noise, target quality, and temporal heterogeneity. The evidence does not separate these explanations. Accordingly, the findings should be interpreted as evidence about the limits of latent temporal compression on the evaluated benchmarks rather than as evidence that temporal structure is absent.

8 LIMITATIONS

The study has several boundaries.

- (1) **Small datasets.** The real datasets contain 15-55 subjects, which limits statistical power and may not represent larger cohorts such as GLOBEM.
- (2) **Three seeds.** All experiments use 3 seeds. This is enough to expose instability but not enough for precise variance estimation.
- (3) **Weak predictive signal in current benchmarks.** All real-data models remain in a narrow AUROC band. This outcome is informative for the characterization objective, but it limits conclusions about downstream utility. Consequently, conclusions about latent temporal compression should be interpreted primarily as characterization results rather than evidence of reliable predictive value in real-world passive-sensing settings.
- (4) **Weak latent identifiability.** Learned representations are not identifiable across seeds and align weakly with synthetic ground-truth factors ($|r| = 0.224$). They should not be interpreted as psychological factors.
- (5) **Incremental-conditioning design.** Because the decoder receives both latent representations and observed traces, the study evaluates the incremental contribution of latent conditioning beyond the observed history. The results therefore do not determine whether latent representations alone contain predictive information that could be exploited under alternative architectures or training objectives.
- (6) **No clinical validation.** The study includes no user study, prospective evaluation, or clinical outcome validation.
- (7) **No deployment evaluation.** The study evaluates predictive and diagnostic behavior only; it does not evaluate downstream interventions, deployment workflows, or clinical decision-making.
- (8) **Incomplete subgroup metadata.** The real datasets do not provide sufficient metadata to support subgroup inference beyond the dataset-specific metadata-derived subgroup calibration diagnostics reported here.
- (9) **Simplified missingness models.** Real data contains structured missingness that MCAR, burst, and fatigue-dependent MNAR stress tests only partially capture.

9 CONCLUSION

This study used latent temporal compression as a diagnostic probe to characterize whether latent conditioning yields reliable incremental predictive gains beyond observed histories under leakage-controlled temporal evaluation. Synthetic data showed that the evaluation design can detect an incremental latent-conditioning benefit when stable temporal dynamics are intentionally introduced, but that benefit did not generalize to Depresjon, StudentLife, or WESAD real-world datasets, where all models remained in a narrow AUROC band. Temporal generalization varied across datasets, calibration worsened under shift, and missingness robustness did not translate into better real-data prediction.

The main scientific takeaway is that the public benchmarks studied here do not provide clear evidence that latent temporal compression yields reliable incremental predictive gains, despite

succeeding in a controlled synthetic setting with known temporal dynamics. The primary contribution is the evaluation design that makes this limitation visible through temporal splits, calibration analysis, missingness stress tests, latent diagnostics, and failure diagnostics rather than through a single random split or aggregate metric. Future work should evaluate larger longitudinal cohorts, more stable behavioral targets, and richer sensing streams using the same temporal-evaluation, calibration, robustness, latent-diagnostic, and failure-analysis setup developed here.

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A DATASET DETAILS

A.1 Depresjon Preprocessing

Original Zenodo dataset (CC BY-NC 4.0): 55 subjects in condition/ (23 depressed) and control/ (32 healthy) directories. Minute-level accelerometer activity counts over 5-18 days. We extract 8 windowed features (mean, std, min, max, energy/RMS, zero-crossing rate, skew, kurtosis) across 48 temporal windows. MADRS scores are from scores.csv for depression prevalence calibration.

A.2 StudentLife Preprocessing

Dartmouth StudentLife dataset: 48 students, 10 weeks of continuous smartphone sensing. We extract 8 behavioral features (conversation duration, activity level, sleep duration, location entropy, etc.) aggregated into 48 temporal windows. Depression labels are from pre/post PHQ-9 assessments.

A.3 WESAD Preprocessing

Accessed via HuggingFace (GeorgiaCh96/WESAD_raw_data, 196,978 rows, 15 subjects). We use 14 HRV features: MEAN_RR, MEDIAN_RR, SDRR, RMSSD, HR, pNN50, VLF, LF, HF, LF/HF, SD1, SD2, KURT, SKEW. Features averaged within 48 time-binned windows per subject. Labels are stress vs. non-stress.

B IMPLEMENTATION AND EVALUATION DETAILS

We use a bidirectional LSTM encoder and a GRU decoder, each with hidden size 64, two layers, and dropout 0.2. The latent head outputs a three-dimensional latent representation ($k = 3$). Models are optimized with AdamW (learning rate 10^{-3}) using a batch size of 16 and early stopping. Patience is set to 15 epochs for the encoder and 10 epochs for the decoder. All reported results are averaged over three random seeds (42-44).

Robustness and calibration are summarized with concise diagnostics. Missingness stress tests use MCAR masking at 10%, 25%, and 50%; burst missingness in contiguous 4-8-step blocks at 25%; and fatigue-dependent MNAR masking with up to 40% masking at maximum fatigue. Calibration is reported with ECE computed using 10 equal-width bins and user-level subgroup ECE derived from dataset metadata. Subgroup labels may vary across datasets and should be interpreted as dataset-specific metadata-derived groups rather than demographic categories. Uncertainty inflation is the ratio of predictive variance under shift to the corresponding held-out-user variance, estimated with Monte Carlo dropout.

All metrics are reported as mean $\pm$ std across 3 seeds, with bootstrap 95% confidence intervals (300 iterations) for the main summaries. We do not report significance tests because the datasets are small and the primary goal is characterization rather than hypothesis testing.

C ETHICAL IMPACT STATEMENT

This paper is a methodological study of longitudinal prediction robustness.

Scope. Synthetic data and three real public datasets with low-AUROC results. No clinical validation. The framework is intended for research use.

Surveillance risk. Temporal representations could intensify monitoring. The evaluated public benchmarks provide limited evidence for reliable individual-level prediction under the evaluation framework studied here.

Subgroup calibration. Dataset-specific metadata-derived group calibration differences appear in the evaluated diagnostics. These differences are descriptive, dataset-specific, and should not be read as verified demographic fairness effects.

Data ethics. All datasets are de-identified and publicly available (CC BY-NC 4.0, Dartmouth archive, HuggingFace).

Dual use. Temporal prediction models could be used for monitoring or prioritization in settings not studied here. Because the evaluated benchmarks provide limited evidence for reliable individual-level prediction, the framework should not be read as supporting decision making without prospective validation.